

Rick Clinton

Independent AI Safety, Governance & Agent Security Researcher

Chennai, India | clinpeter@gmail.com | github.com/Rick-Clinton-jpg | Research portfolio available on request

RESEARCH PROFILE

Independent researcher studying how deterministic, externally verifiable controls can constrain AI reasoning and agent action. Since July 2026, I have directed an AI-assisted portfolio spanning agent authorization, multi-agent delegation, trajectory monitoring, adversarial evaluation, and auditability. I define the threat models, architectures, invariants, experiments, and acceptance criteria; coding agents assist with implementation, testing, and documentation. Earlier, I spent 5+ years delivering quality- and SLA-driven programs at Walmart and Amazon.

RESEARCH FOCUS

AI control and monitoring, agent security, deterministic governance, threat modelling, red-team and mutation evaluation, capability-based execution, multi-agent delegation, research writing.

SELECTED AI SAFETY RESEARCH

Mirage | github.com/Rick-Clinton-jpg/Mirage | v0.5.2, September 2026

- Directed a three-zone Linux containment experiment treating an AI evaluation sandbox's repository mediator as compromised; 23 of 23 required controls passed and the permitted package path succeeded in 10 of 10 fresh topology trials, while arbitrary destinations, alternate ports, redirect follow-up, and a reverse pivot into management stayed blocked.
- Adversarial review broke the original hash-chain verifier by rewriting and rehashing a failed result into an authenticated pass. Removed it as the default compliance path, required witnessed authentication, and kept the exact forgery as a permanent regression test.
- Exported offline-verifiable evidence (canonical JSONL, witness receipt, signed policy and vulnerability assurance inputs, SHA-256 manifest) so a reviewer can check the result without trusting the originating lab.

Mirror World | v0.8.9.9, September 2026

- Ran adversarial escape campaigns against real free-tier LLMs inside a kernel-hardened sandbox with real seccomp and Linux namespaces; zero real capability grants across every trace event, including against a model that independently discovered a five-step forged-authority escalation chain on its own initiative.
- Found and fixed a timing side-channel that let an attacker distinguish real operations from deception-layer reflections with 99.2 percent classification accuracy purely from response latency, defeating the system's core indistinguishability premise. Shipped a constant-time fix, then verified it correctly: an initial 61.3 percent in-sample re-test looked promising, but a held-out, cross-validated test showed the true residual signal was 52 to 55 percent, consistent with chance.
- Caught the regression test meant to guard that fix in the same trap it was built to prevent, a single train and test split swinging between 0.52 and 0.66 on unchanged code, and corrected it by averaging over 20 independent splits.

Armour | github.com/Rick-Clinton-jpg/Armour | August–September 2026

- Directed a deterministic policy and execution boundary for AI agents: an agent proposes an action; Armour independently authorizes, rejects, or escalates it before a registered handler runs. The project explicitly remains a gate, not a sandbox.
- Hardened production construction to require a trusted isolated Ed25519-or-equivalent approval verifier and a durable, integrity-protected approval ledger; added replay protection, bounded security memory, hash-chained receipts, and optional deletion-detection checkpoints.
- Added single-use filesystem and network capabilities that bind the resource checked to the resource supplied to the handler, reducing destination-substitution races. Public main: 146 passing tests and 20 of 20 modeled mutants killed; no real handler executes during mutation runs.
- Kept a higher-risk deception experiment off the main branch after self-adversarial probes exposed three weaknesses. Added non-negotiable ceilings, control-character-safe reflections, strict input types, and a blind-review protocol; local branch reaches 165 tests and 27 of 27 mutants, but is not claimed as independently verified.

Chainmail | github.com/Rick-Clinton-jpg/Chainmail | v5.1, September 2026

- Directed an independently hardened governance layer for long-running multi-agent systems around one invariant: an agent can never delegate more authority than it currently holds. Human-declared authority is the ceiling; contextual signals may only tighten decisions.

- Made nonce and proposal replay state, restrictions, delegated authority, permission budgets, and fleet and per-agent step budgets durable and atomic in SQLite. Production mode fails closed without signatures, durable state, and a non-permissive execution boundary.
- Expanded adversarial evaluation with an authority-laundering family covering over-delegation, stale authority after revocation, forged budget fields, multi-hop laundering, and concurrent double-spend. Fresh local rerun: 203 passed, 8 skipped.

Moss | *self-healing sandbox containment, 2026*

- Directed a biologically-inspired containment architecture where a central supervisor coordinates peripheral defensive units that detect and quarantine hostile processes already inside a sandbox, without acting outside the sandbox boundary. Built a working prototype with HMAC-signed containment modules and fail-closed capability removal, then extended it with a Linux enforcement adapter (namespaces, cgroups v2, seccomp) on a disposable, isolated virtual machine.

APPLIED SYSTEMS

Buy or Wait? | *HackerRank Orchestrate, September 2026*

- Built a dual-mode financial affordability agent (local deterministic engine by default, opt-in API mode) under a 21-hour deadline; added a contradiction check that flagged ambiguous income evidence rather than trusting it, deliberately lowering categorical accuracy from 20 of 25 to 18 of 25 to avoid recommending payment plans against unverified income.
- Diagnosed the tradeoff by isolating each safety check's individual effect on scoring, then kept the stricter, lower-scoring configuration, prioritizing safety over an optimal benchmark score. Ranked 937 of 3,062; certificate verified by HackerRank.

Quorum | *Google All Things Agentic Hackathon*

- Directed deployment of a gated multi-agent coordinator on Google Cloud Run: a Gemini and Vertex AI worker drafts patches, while three independent deterministic verifiers must agree before a patch can proceed; stages are written to a Firestore audit trail.
- Converted a cross-instance Firestore race, a regex denial-of-service risk, and an unbounded retry loop into reproducing regression tests and fixes; the system opened real, verifiable pull requests only after its checks passed.

SELECTED WRITING

JUDGE | *position paper, 2026*

Proposes a five-layer architecture that converts model behavior into explicit claims and deterministic, externally replayable gates; frames falsifiable comparisons among self-report, model judging, and external verification.

Building With a Conditionally Trusted Machine | *working paper, 2026*

A first-person account of using AI without granting it unearned authority, including the implementation knowledge debt created by AI-assisted development and the controls built in response.

The Five-Layer Stack | *working paper, August 2026*

Develops a deterministic external-verification stack from observed failure modes and uses invented-entity cases to reduce prior-knowledge confounds.

PROFESSIONAL EXPERIENCE

Project Management Specialist — Walmart, Chennai | *August 2022–June 2025*

- Directed an eight-week onboarding program for 15 to 22 international sellers per month across three years, serving 700+ clients while maintaining 100 percent customer satisfaction; used discovery research to tailor each engagement.
- Served as the primary escalation point, trained the broader business-development team, and designed a duplicate-item identification playbook that remains in use; received Stellar Stakeholder Recognition, April 2023.

Catalogue Associate — Amazon Pvt Ltd | *April 2019–May 2021*

- Managed high-volume catalog operations and vendor coordination against defined SLA and quality standards.

EDUCATION

B.Tech, Mechanical Engineering — SRM University

Diploma, Modern Applied Psychology (DiMAP)